Type formers as maps out of classifying presheaves: a more modular presentation of the model

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1 The situation

In `seg/model` each construction is made on an arbitrary context and its stability under substitution is then proved for that construction: El along a code map (`GElC`), Ext(α) and its fibration (Theorem 7, `GExtSub*`, `GExtNat*`), the glued type and its classifier (`GUaOp6`, `GUaOp8`), the side data of the composition of the universe (`GCompU2`, `GCompU3`). Each such proof compares two constructions that agree only up to equations, and transports the structures (compositions, fillers, equivalences, classifiers) along them. Measured over the 88 files of the univalence development (about 9800 lines): 36% is the mathematics proper, 39% proves that substitution commutes with a construction, 15% transports structures along equations of types, 8% moves structures between universe levels.

2 The idea

A type former is defined *once*, on a presheaf that classifies its data, and every instance is obtained from it by substitution. This is the point of view of natural models (Awodey) and of universe categories (Voevodsky), and it is how local universes (Lumsdaine–Warren) obtain type formers that are strictly stable under substitution.

El. The universe $\mathcal{U}$ is a presheaf of codes, and El is one type over $\mathcal{U}$, with small fibres $\mathrm{El}(F)$. A family given by a code map $c : \Gamma \to \mathcal{U}$ is $\mathrm{El}[c]$, the substitution of El along c. Then

$$\mathrm{El}[c][s] = \mathrm{El}[c \circ s]$$

is the composition law of substitution, proved once for all types, and not a property of El. The fibration of $\mathrm{El}[c]$ is the fibration of El substituted along c; its stability is again the composition law, now for fibrations, proved once.

Ext. The type $\mathrm{Ext}(\alpha)$ at a point ρ of Γ depends only on the data of α restricted to ρ (we already use this "local-data form"). Let $\mathcal{L}$ be the presheaf of such local data: at stage I, a fibrant type A over I, a cofibration ψ, a fibrant type B over the restriction to ψ, a map $\alpha : B \to A$ and an equivalence structure on it. Define $\mathrm{Ext}_{\mathcal{U}}$ once as a type over $\mathcal{L}$, with its fibration (Theorem 7, once), and for data α over Γ with local data map $\ell_\alpha : \Gamma \to \mathcal{L}$,

$$\mathrm{Ext}(\alpha) := \mathrm{Ext}_{\mathcal{U}}[\ell_\alpha].$$

Stability under $s : \Delta \to \Gamma$ then reduces to two facts:

$$\mathrm{Ext}_{\mathcal{U}}[\ell_\alpha][s] = \mathrm{Ext}_{\mathcal{U}}[\ell_\alpha \circ s] \qquad \text{and} \qquad \ell_{\alpha[s]} = \ell_\alpha \circ s.$$

The first is the composition law, once for all types. The second is one equation between data (the restriction of the data of $\alpha[s]$ to a point is the restriction of the data of α to its image), the only thing left to prove for Ext. The same applies to the glued type, to the data of the univalence operation (already a presheaf of data, `GUnivEq.UaU.Q4`), and to the side data of the composition of $\mathcal{U}$.

Levels. $\mathcal{U}$ is a presheaf at level $a{+}1$ whose types have fibres at level a. The operation needed is substitution of a *small* type over a *large* context along a map from a small context. Defined once, it takes the place of the present transfer layers, which move each structure down a level separately.

3 What has been checked

On a copy of the model, without changing the type theory:

- `GSmall`: small types over large contexts, maps into them, and substitution along these maps, with the composition law proved once for all small types (22 declarations).
- `GSmallEl`: El defined once as a small type over $\mathcal{U}$. The present El along a code map (`GElC.Fam`) is *by definition* El substituted along it (checked by `refl`), and the present lemma `elSub`, with its exact statement, is an instance of the general composition law.

Not yet checked: the same for fibrations (mechanical), and Ext over a presheaf of local data, which needs a coding of $\mathcal{L}$ at level $a{+}1$ as for $\mathcal{U}$.

4 What the presentation would look like

1. The base: a category, cofibrations, an interval with connections.
2. A universe: a presheaf of codes $\mathcal{U}$ with El fibrant.
3. For each type former F: a presheaf $\mathcal{L}_F$ classifying its data, a type $F_{\mathcal{U}}$ over $\mathcal{L}_F$ with its structure (for Ext: Theorem 7), and the equation $\ell_{\alpha[s]} = \ell_\alpha \circ s$.
4. Generic substitution laws, once: for types, for fibrations, for equivalence data.

The results stay abstract (any base satisfying (1)); what disappears is the proof, for each construction, that it commutes with substitution.

Remark. This is the semantic counterpart of Ehrhard's presentation of categories with families compared with Dybjer's: there the lifted substitution is taken primitive so that composite substitutions coincide by construction; here each type former is taken primitive on its classifying presheaf so that its instances coincide by construction, and only the equations of data remain.